Multimedia Signal processing

Prof. Marco Tagliasacchi

July, 3 2013

1. **Question (8 pts)**:
   Consider a sequence of binary simbols *aabaaabaaaaabbaaaaba* emitted by a source X.
   (a) Learn the p.m.f. of the source from the observed data and compute the entropy of the source (assuming a memoryless source).
   (b) Learn the conditional p.m.f. $Pr(X_n|X_{n-1})$.
   (c) Estimate the number of bits needed to encode the sequence *aaaba* using Elias coding in two cases: i) without exploiting source memory; ii) exploiting source memory.
2. **Question (5 pts)**
   (a) Illustrate the architecture of motion-compensated predictive video coding.
   (b) Discuss the need for a closed-loop architecture.
3. **Question (5 pts)**
   Illustrate the KLT transform, its design principles, optimality properties and drawbacks.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**
   (a) Consider an audio coding system
   - [T] [F] Perceptual audio coding can be used to encode high-quality speech.
   - [T] [F] Quantization is applied directly in the time domain.
   - [T] [F] The MDCT achieves perfect reconstruction when the length of the analysis window is kept constant.
   - [T] [F] The perceptual model is encoder-dependent.

   (b) Consider a discrete source
   - [T] [F] $L_S(x) \geq L_H(x)$, where $L_S(x)$ and $L_H(x)$ denote, respectively, the average code length of Shannon and Huffman coding for a discrete memoriless source x
   - [T] [F] $L_S(x) = L_H(x)$, if and only if the p.m.f. is dyadic.
   - [T] [F] The code length of each codeword is upper bounded by $H(x) + 1$

   (c) Consider a quantization scheme based on compander-expander.
   - [T] [F] The quantization symbols are equally likely.
   - [T] [F] The mean square error distortion is less than that achievable by the Lloyd-Max quantizer, for the same number of reconstruction levels.
   - [T] [F] The input source x is mapped to an intermediate source y by means of the mapping $y = x^{\alpha}$, $\alpha \in \mathbb{R}$

5. **Question (Matlab - 10pts)**

   Image tampering detection and localization can be performed exploiting the traces left by JPEG compression.

   A forensic analyst wants to analize image *image.png*, which is available in a lossless compression format.

   (a) Determine whether the image was previously compressed with JPEG. To this end, perform the block-wise 2D-DCT transform, with blocks of size 8×8, and analyze the histograms of DCT coefficients. Comment your results.

   (b) Determine (by visual inspection) the quantization step sizes used for the DCT coefficients $(2, 3)$ and $(4, 4)$.

   (c) Shift the image by 1 pixel both horizontally and vertically. Repeat the previous steps on the shifted image. Comment your results.

   (d) As an alternative method, compress the image *image.png* using different JPEG quality factors QF (Hint: use `imwrite(Image, filename, 'Quality', q)`). Compute the PSNR between the image *image.png* and the compressed images. Plot the curve PSNR vs QF and comment your results.